HMG Sequencing Workshop

Module 2 computer laboratory

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1 Introduction

Module 2 computer laboratory

Intended Learning Outcomes

By the end of this practical, you should be able to:

- navigate aligned short-read sequencing data in IGV
- interpret genes, exons, introns, reading frames, repeats and regulatory annotations
- identify candidate sequence variants and distinguish them from likely sequencing or alignment artefacts
- assess Mendelian inheritance in a parent-child trio
- phase nearby variants into haplotypes using family information and paired reads
- investigate population frequency and predicted functional consequences using Ensembl, VEP and gnomAD

The advent of short-read sequencing technologies, allowing high-throughput DNA sequencing at relatively low cost, has revolutionised genetics over the last fifteen years or so. In particular the ability to sequence whole human exomes, genomes and transcriptomes, has transformed how we conduct human genetics, yielding radical new approaches to human disease gene identification and allowing us to ask questions largely unimaginable just a few years ago. However, although in many areas it is relatively straightforward to generate DNA sequence data, interpreting the masses of data generated remains a major logistical and bioinformatic challenge. Fortunately, data analysis and interpretation are still largely based on fundamental genetic principles and our knowledge of the molecular pathway from DNA to RNA to protein.

In this session we will familiarise ourselves with a web app version of the Integrated Genome Viewer (IGV). This app allows us to visualise aligned NGS data, to identify sequence variants and to sort neighbouring variants into haplotypes. To do this we will use publicly accessible whole genome sequence data from a parent-child trio derived from the 'Platinum genomes', described in:

- Eberle, M.A., Fritzilas, E., Krusche, P., Kallberg, M., Moore, B.L., Bekritsky, M.A., Iqbal, Z., Chuang, H.Y., Humphray, S.J., Halpern, A.L., Kruglyak, S., Margulies, E.H.,

McVean, G. and Bentley, D.R. (2017) A reference data set of 5.4 million phased human variants validated by genetic inheritance from sequencing a three-generation 17-member pedigree. *Genome Res. 27, 157-164.* https://doi.org/10.1101/gr.210500.116

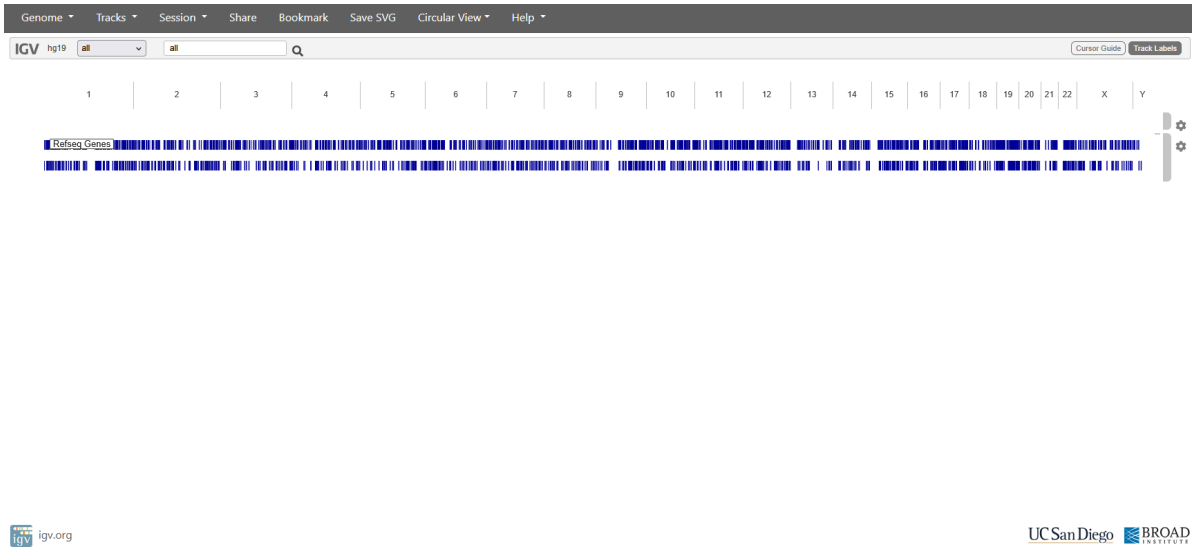
These short-read data were obtained using whole genome sequencing via paired-end 2x100 bp reads on an Illumina HiSeq2000 platform. Sequencing was performed to an average depth of 50x.

2 Set up IGV and annotation tracks

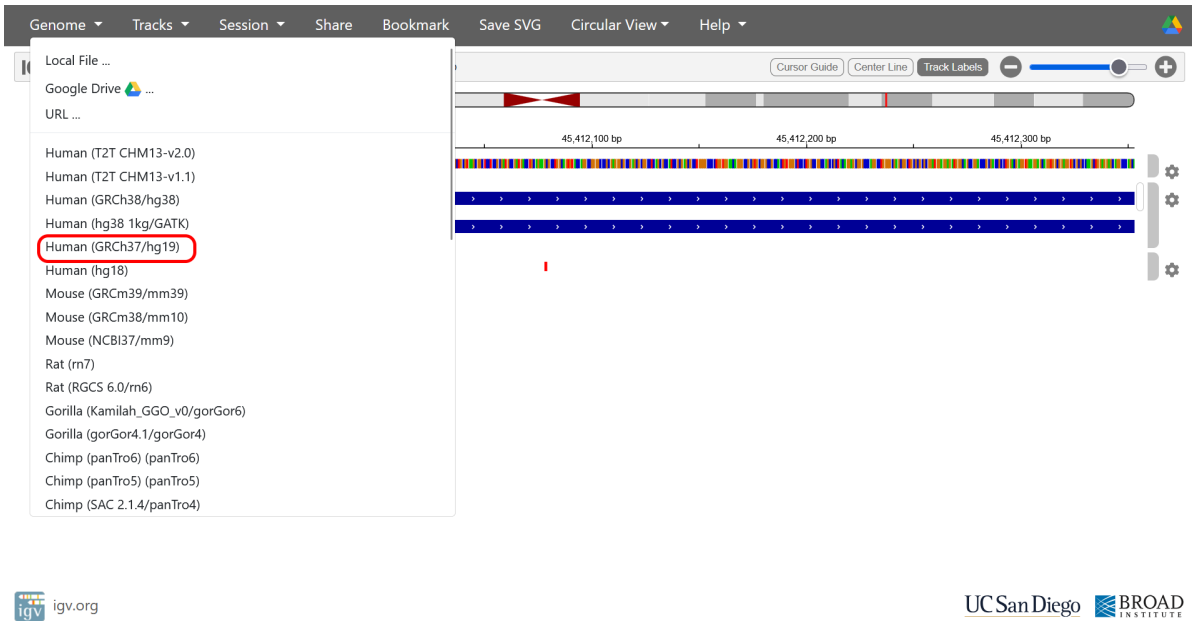
Use your favourite internet browser to navigate to https://smb-informatics.mvls.gla.ac.uk/igv

(Note: if off campus, you will need to connect using the University VPN - https://www.gla.ac.uk/myglasgow/it/vpn/)

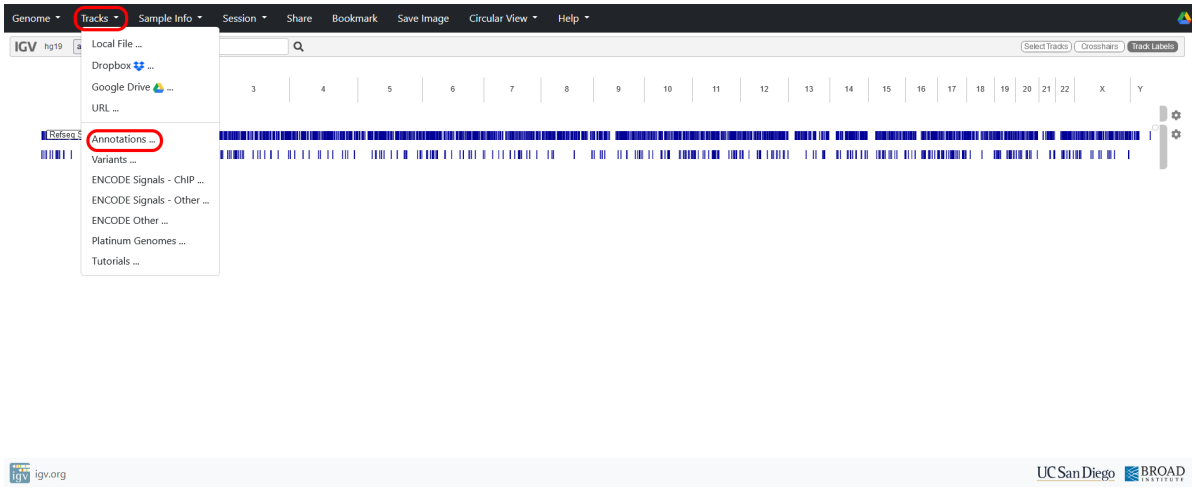
You should see a window like this:



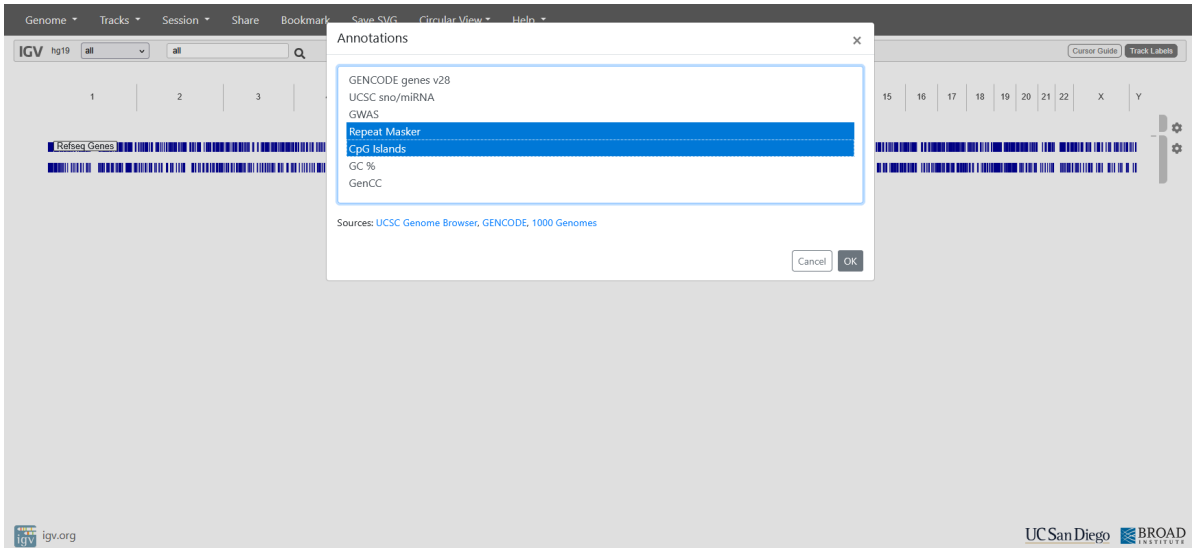
The platinum WGS data we will be using have been aligned to version hg19 of the human reference genome, so we will need to first select the GRCh37/hg19 reference sequence from the genome menu:



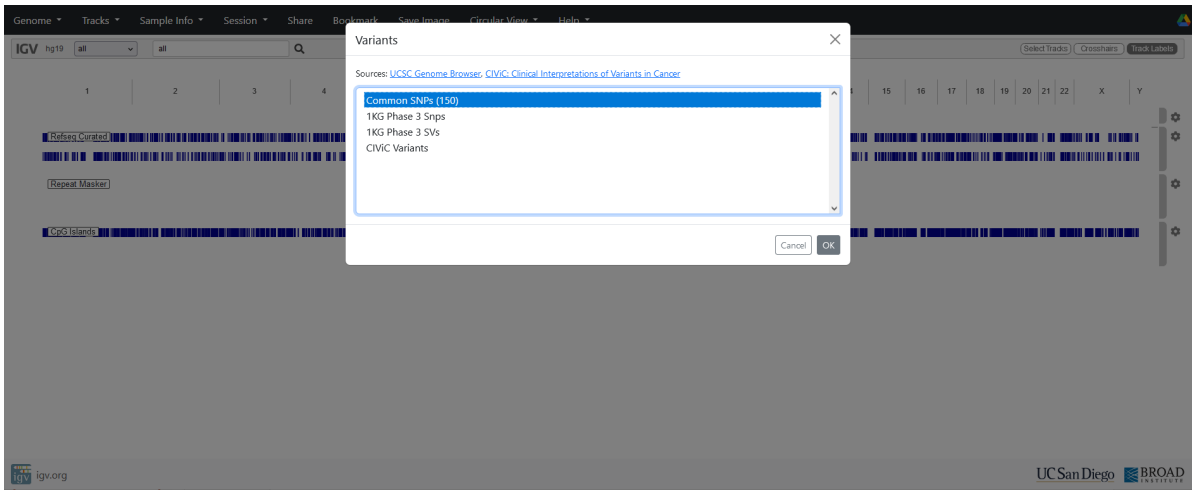
In order to interpret the sequence data, we are going to add some additional "Annotation" "Tracks" to the IGV window:



Use Shift+Click to select the Repeat Masker and CpG Islands annotation tracks (in addition to the Refseq Genes track which is a default track):



We can also add some known "variants" as a track from the dbSNP database (version 150):



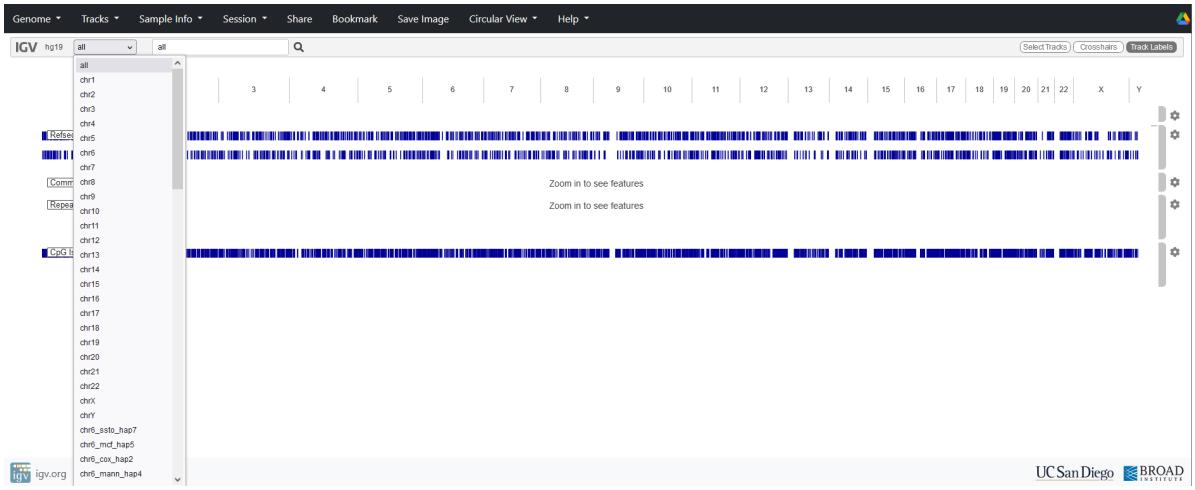
Your window should now look something like this (you can use the grey handle on the right to drag the Refseq Gene annotation back to the top):


Genome
Tracks
Session
Share
Bookmark
Save SVG
Circular View
Help
IGV
hg19
all
all
Cursor Guide
Track Labels
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
X
Y
Refseq Genes
Common SNPs (150)
Zoom in to see features
Repeat Masker
Zoom in to see features
CpG Islands
igv.org
UC San Diego
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3 Navigate the genome

Before looking at any sequence data, get a feel for navigating around the genome and remind yourself of some features of genome organisation.

To see more detail, we need to focus on a smaller region of the genome. There are various ways to do this, including selecting a chromosome either using the pull-down menu, or the chromosome buttons:



Let's look at chromosome 8:



At this level we can start to see some detail regarding the distribution of known genes (A) and CpG islands (B).

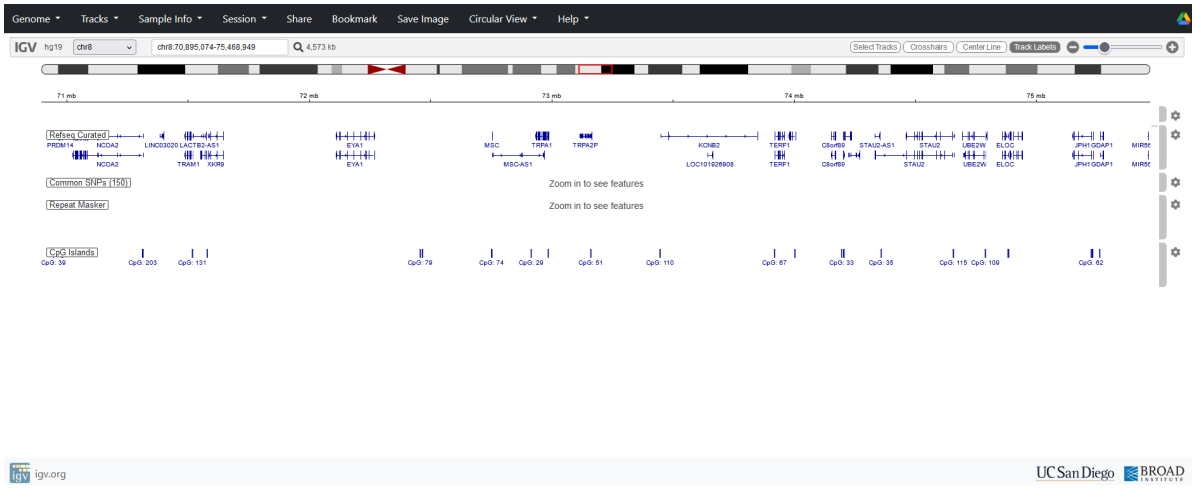


We can zoom in further by either clicking and dragging on the scale (A), typing in sequence coordinates (B), or using the zoom buttons (C).

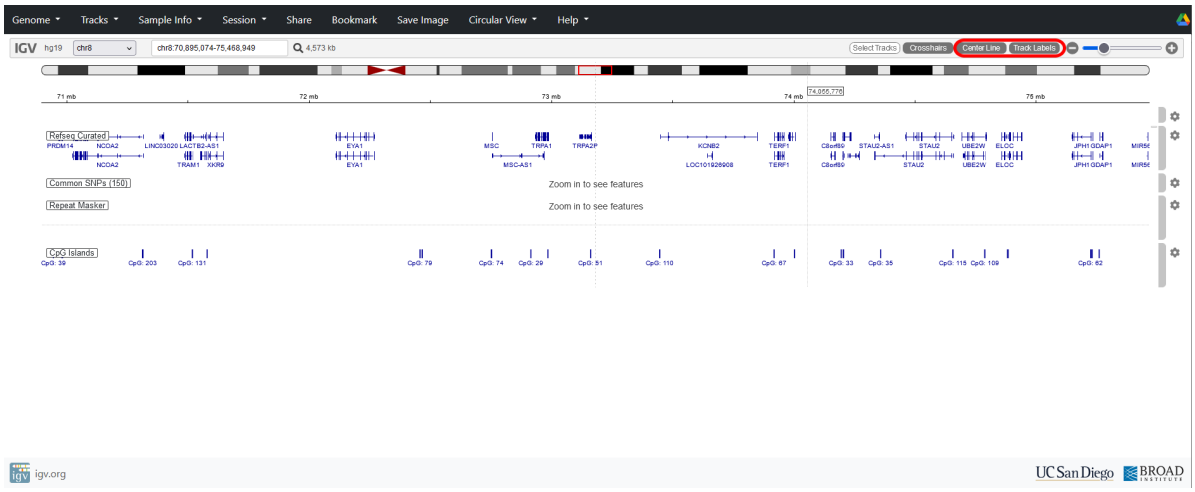


Try pressing the + zoom button **five** times:

You should now start to see individual genes and CpG islands. Note how genes and intergenic regions differ in size.

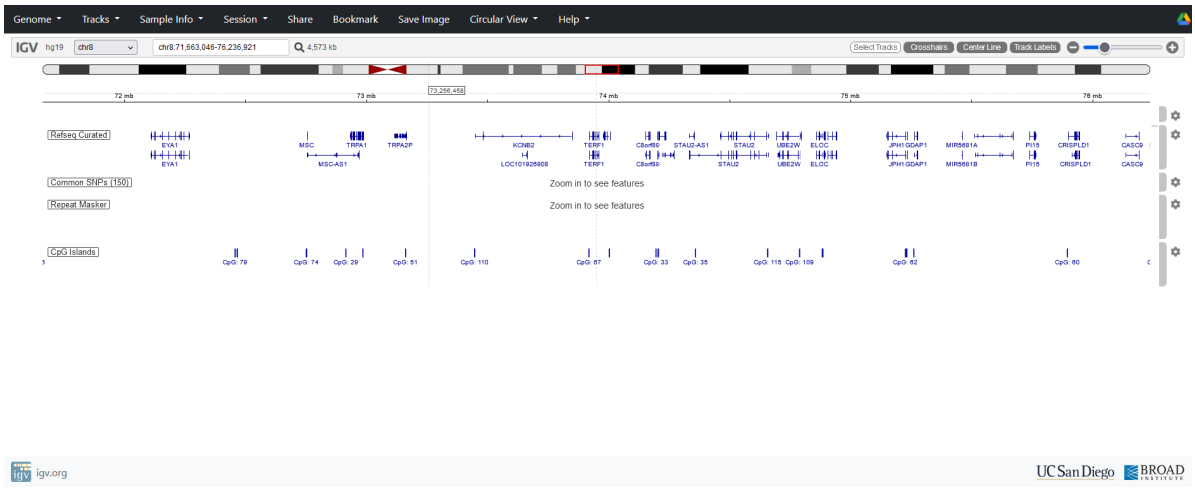


You can press the “Cursor Guide” and “Center Line” buttons to help with navigation.



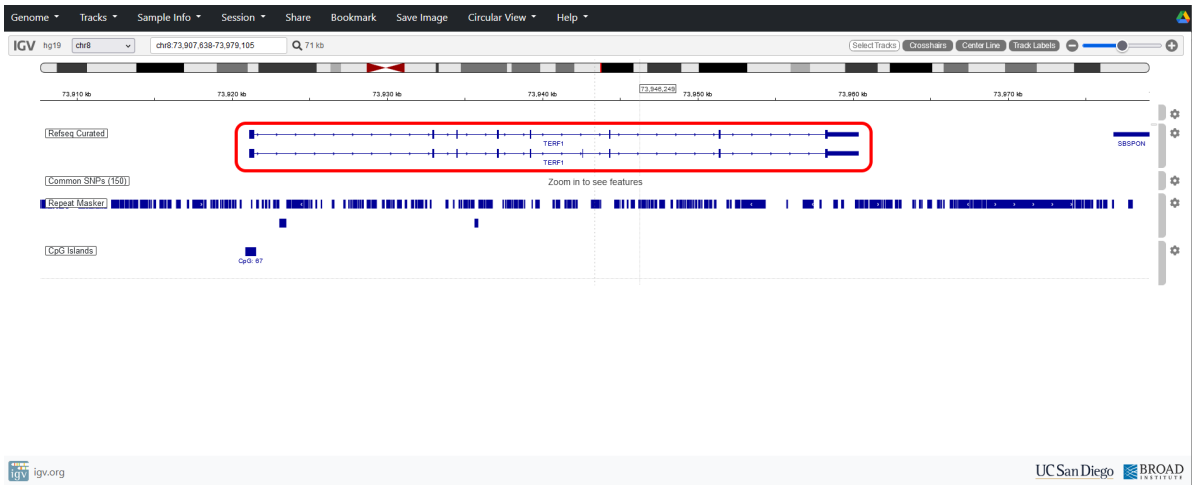
4 Interpret gene structure using *TERF1*

Once you have zoomed in a bit, you can now also navigate left and right by clicking and dragging the cursor in the data window. Move the *TERF1* gene to the centre of the frame:



Exons are indicated by the dark blue boxes with coding regions taller than the non-coding parts of the exons. **Introns** are indicated by the thin blue line, with the direction of transcription indicated by small arrows (>).

Zoom in to see the structure of the *TERF1* gene (there are two primary transcripts for this gene):



Test your understanding

- How many exons does the *TERF1* gene have?

Answer

TERF1 has two primary transcripts, one with 9 exons (upper) and one with 10 (lower).

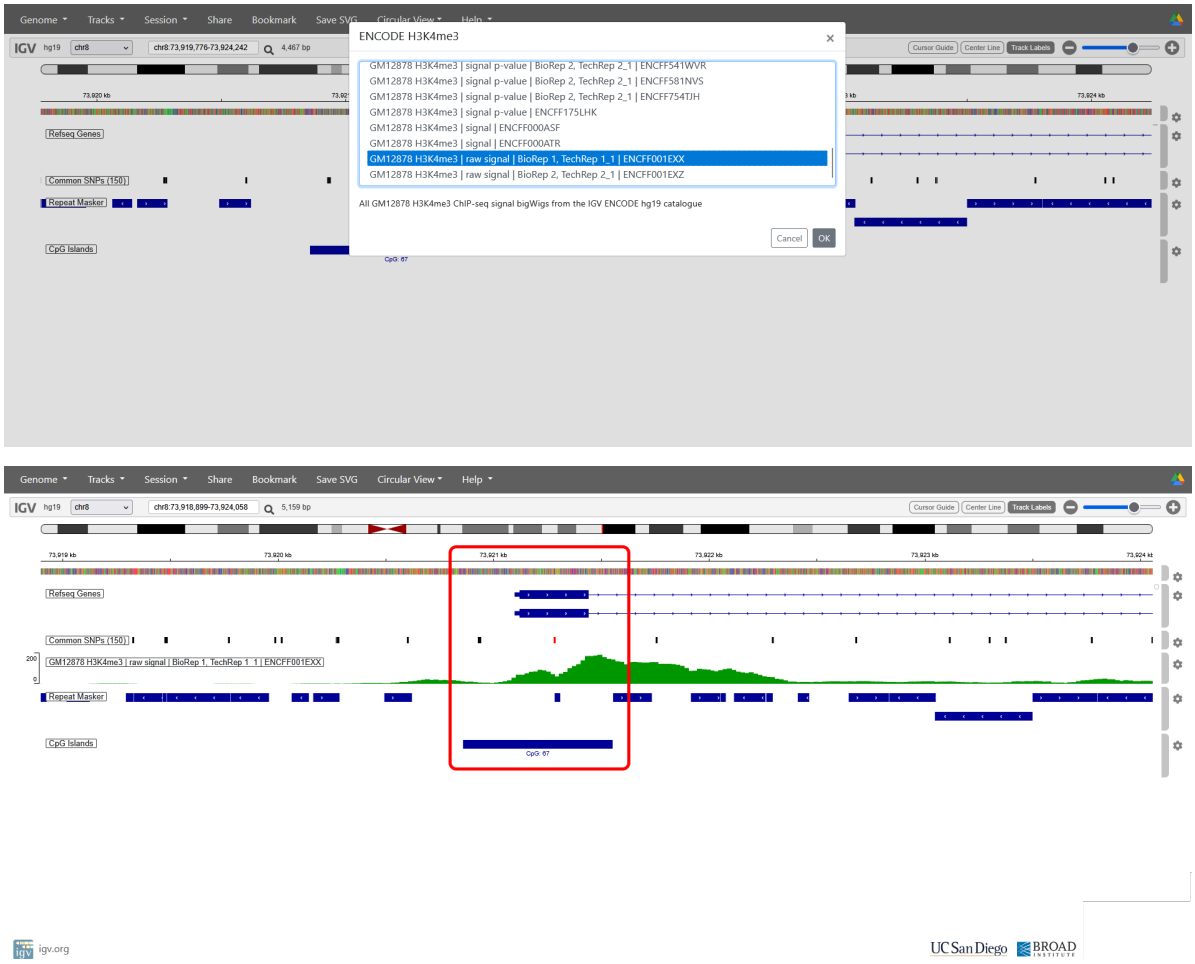
Move exon 1 of *TERF1* to the centre and zoom in some more. You should start to see the reference sequence appearing. You can also press the cog on the track to turn on "Three-frame translation" to aid in interpretation. This shows the amino acid sequences corresponding to the three possible reading frames of the DNA sequence.



4.1 Add promoter-associated chromatin data

We can add some additional information about the accessibility of DNA in this region by adding an ENCODE Signal track containing ChIP-seq data for H3K4me3 (trimethylation of lysine 4 on histone H3). This histone variant is associated with open chromatin and marks active promoters in the genome, typically appearing near transcription start sites (TSSs).

Add the ENCODE H3K4me3 track and select the 'GM12878 H3K4me3 | Raw Signal | Biorep 1' experiment near the bottom of the list and press OK to add the track.



igv.org

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Test your understanding

- Is CpG island 67 positioned where you expect it to? (zooming in might help)
- Why?
- Which of the three reading frames do you think is correct for exon 1?

- What is the name of the first repeated element in intron 1?

Answer

- Yes, CpG island 67 spans a promoter region in exon 1 and associates with H3K4me3 ChIP-seq data (typically found at TSSs and in open chromatin).
- The correct frame starts with an ATG and has no STOP codons within exon 1. It will depend on your particular instance of IGV.
- The first repeated element is a MIRb element and there are some additional Alu, LINE and MIR elements, and simple sequence repeats, further down. Note that ~80% of intron 1 sequence is some sort of repeat.

4.2 Examine splice junctions

Splice junctions mark the boundaries between exons and introns and are critical for the accurate removal of intronic sequence during RNA splicing. Adjust the view to examine the end of the first exon.

Warning

Be careful - the vertical order of the reading frames sometimes jumps about as you zoom in and out.



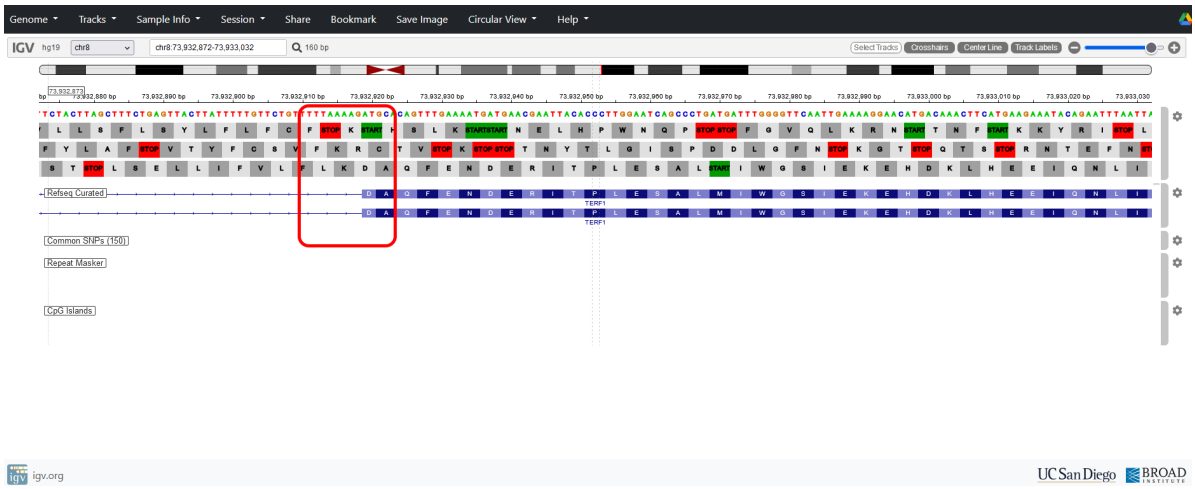
Test your understanding

- What is the sequence of the last codon of exon 1?
- What is the sequence of the first two bases of the intron?
- Why is this important?

Answer

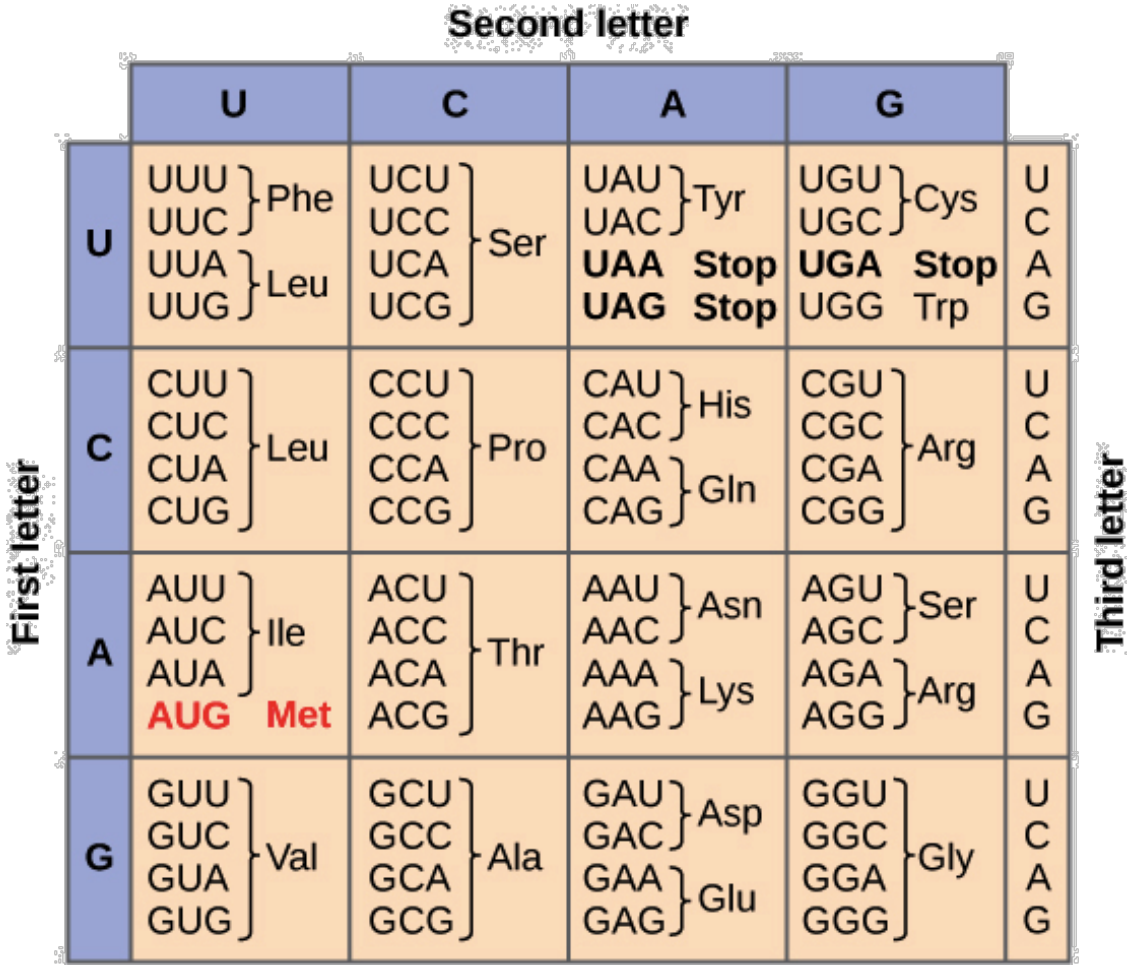
- It will depend on your IGV instance but in this image it is the bottom frame as this matches the start codon at the beginning of the transcript, so the last codon is **G**–
- The first two bases of the intron are GT
- This is the consensus splice donor site

Adjust the view to examine the end of the first intron and the beginning of the second exon.



Test your understanding

- What is the sequence of the first codon of exon 2?
- What codon is formed after splicing when the final base of exon 1 is combined with the first two bases of exon 2?
- Which amino acid does it encode (see below for the genetic code)?
- What is the sequence of the last two bases of the intron?
- Why is this important?



Answer

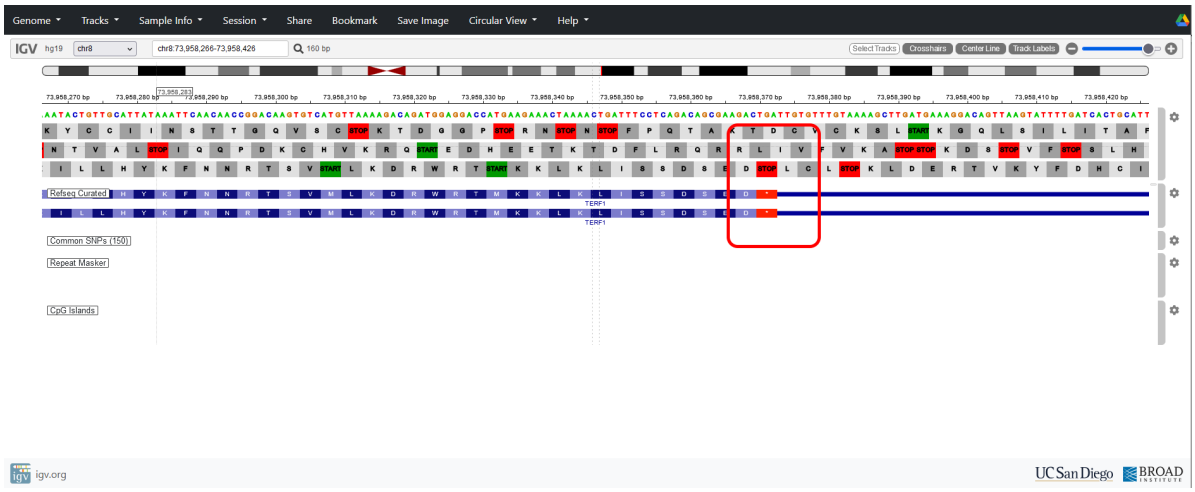
- The codon so far contains **G**– and was split by the intron. To make it a complete triplet, the rest of the codon must be the -*AT* from exon 2.
- GAT.
- Aspartic Acid (D)
- The last two bases are AG
- This is the consensus splice acceptor site.

Scroll through the gene and note how the intronic sequences are comprised largely of dispersed repeat sequences.



4.3 Examine the terminal exon and 3' UTR

Navigate to the terminal *TERF1* exon and zoom in:



Test your understanding

- Which is the correct reading frame for this exon in your view?
- What is the sequence of the stop codon?

Answer

- It will depend on your IGV instance but in this image it is the bottom frame as the stop codon coincides with the end of the terminal exon.
- TGA

Zoom out to see the whole of the final exon.



Test your understanding

- What is the approximate length of the 3' UTR for *TERF1*?

Answer

The UTR spans 73,960,358 to 73,958,372 so is 1,986 bases (approx. 2,000 bp)

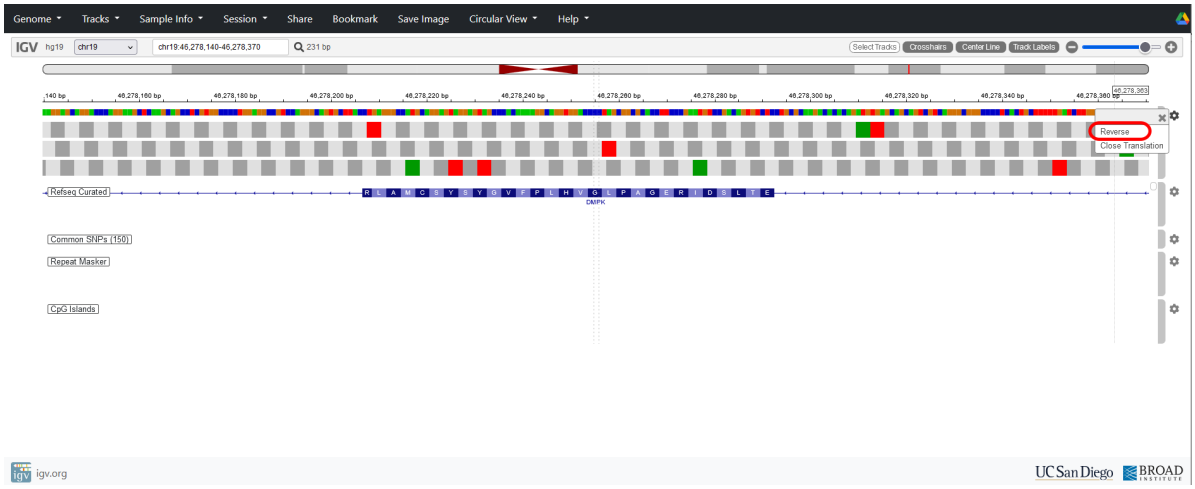
5 Genes on the reverse strand: *DMPK*

Let's look at the *DMPK* gene. This gene is different from *TERF1* as it is transcribed from the bottom strand relative to the chromosomal reference (the top strand). Exon 1 is on the right, and the terminal exon on the left. You can quickly navigate to *DMPK* (or any gene) by typing its name into the search bar.



Zoom in on one of the DMPK coding exons. To see the correct amino acid sequence for this gene, we will have to set the 'reverse' option so that IGV translates from the bottom strand. Remember that it will still be running right to left.

> **Warning**
>
> Don't forget to turn this option off again if you look at a gene or variants on the reference (top) strand.



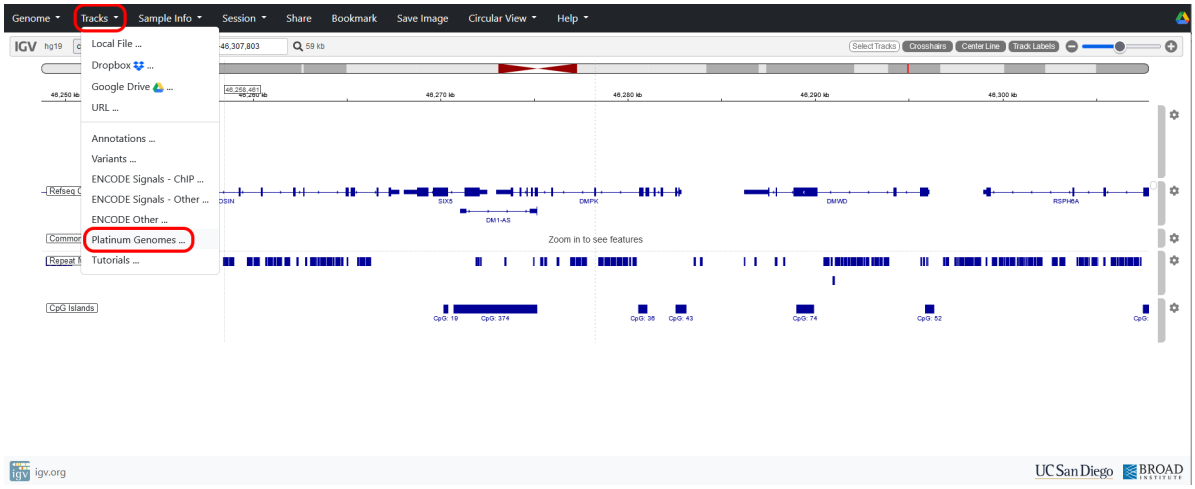
Zoom out a bit more to get an overview of the other genes in the vicinity of *DMPK*. Some genes are in the same orientation as DMPK and transcribed from the bottom strand (e.g. *SIX5*, *DMWD*, *RSPH6A*) but others are transcribed from the top strand (e.g. *BHMG1*).

You should also see that many genes have a CpG island spanning exon 1. These regions are often (but not always) associated with the presence of H3K4me3 in ChIP-seq data, indicating open chromatin.

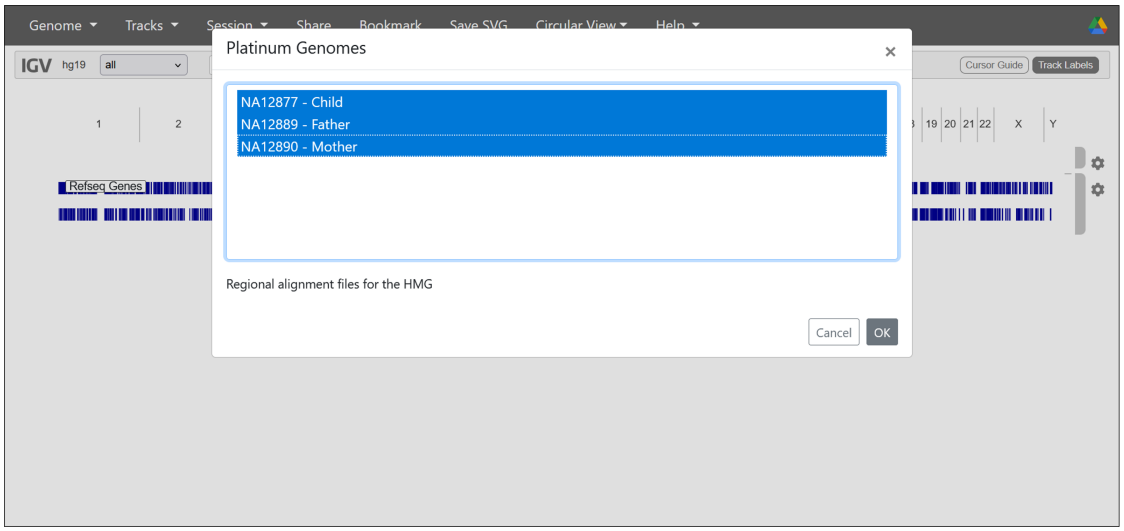


6 Load Platinum Genomes sequencing data

Next, we will load some actual sequence data from the platinum genomes collection of tracks:



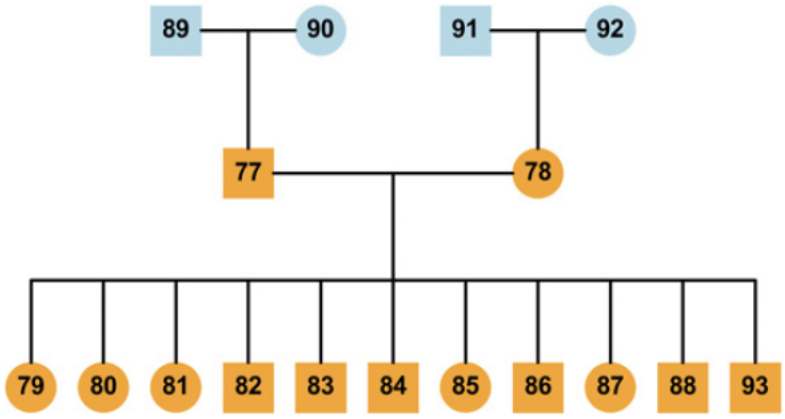
Shift-click samples **NA12889**, **NA12890** and **NA12877** and press OK.



The platinum genomes are derived from a reference genome dataset described in Eberle *et al., 2017*. In the study, the genomes of four grandparents, two parents and 11 children were

sequenced on an Illumina HiSeq2000 to an average depth of 50× using 2×100 bp paired-end reads.

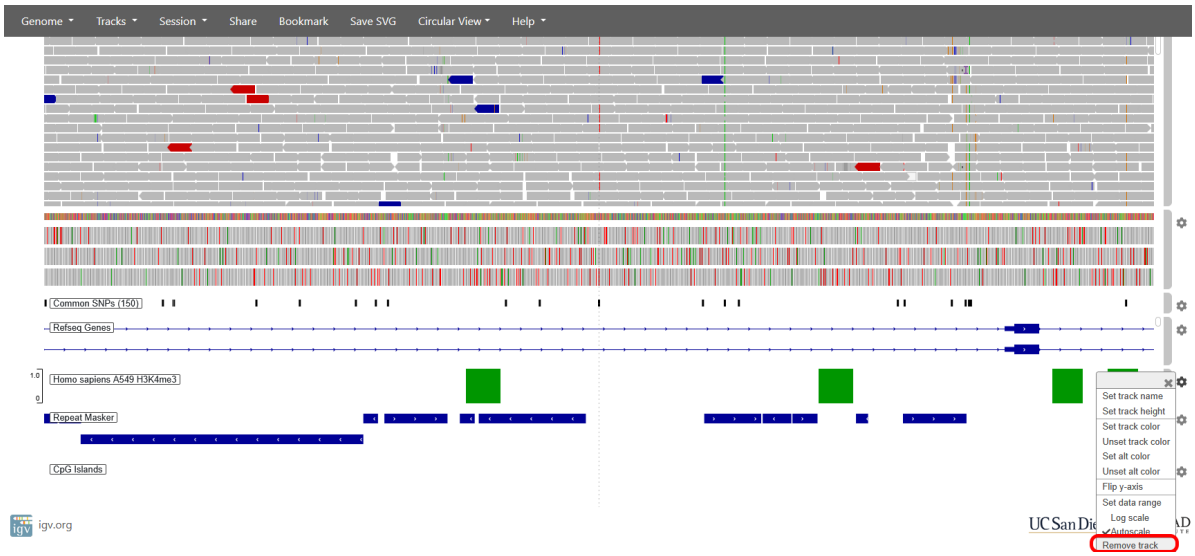
The ID of each sample is defined by adding the family identifier prefix (e.g. NA128) to each numbered individual (e.g. 77 would be NA12877). For this session, we will load in a mother-father-child family trio (individuals 89, 90 and 77) from family NA128.



You may need to use the cog associated with each track to change the Track Height to get everything visible on your screen. You can also drag the tracks into an easier order using the handles at the side of each track.



We can speed up the display by removing tracks we no longer need (e.g. CpG, repeats and H3K4me3) by clicking their specific cog on the track and selecting remove.



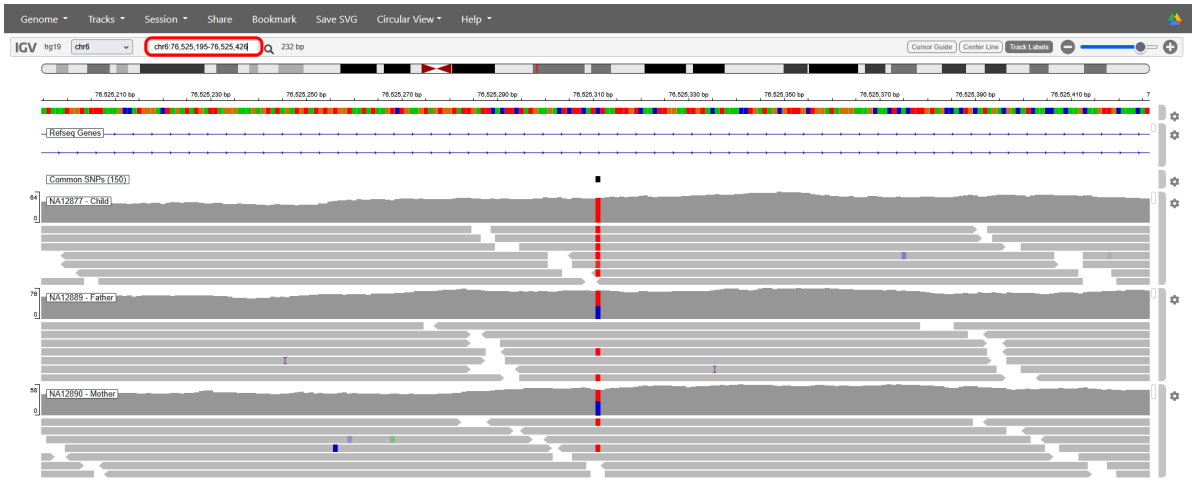
> **Warning**
>
> Sometimes IGV still slows down when you load the sequence data and you may need to refresh the page if it gets stuck. Unfortunately, you will then need to change the genome to hg19 again and reload the relevant tracks again.

7 Inspect reads and identify variants in *MYO6*

Let's zoom in and have a look at a region in the *MYO6* gene of the trio in detail by typing the chromosome coordinates into the search box: chr6:76,525,195-76,525,426.

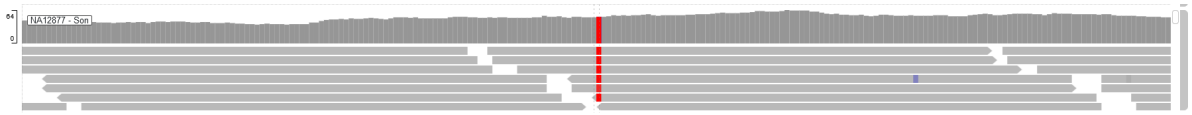
MYO6 mutations are linked to autosomal dominant and autosomal recessive inherited hearing loss.

> **Warning**
>
> We will be examining variants in the top reference strand, so it is best to turn off the 'reverse' option for this region unless looking at whether a variant is synonymous

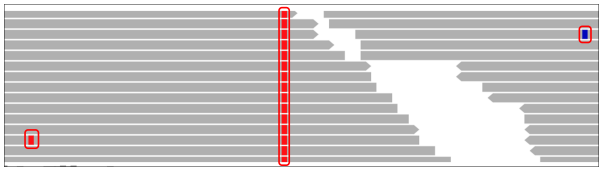


The mountain plot at the top of the display shows the read depth across the region (the scale here goes from 0 to 52x). Individual 100 bp reads are shown as grey rectangles - the arrow on the end indicating the direction of the read (you can use the scroll bar here to see more of the reads).

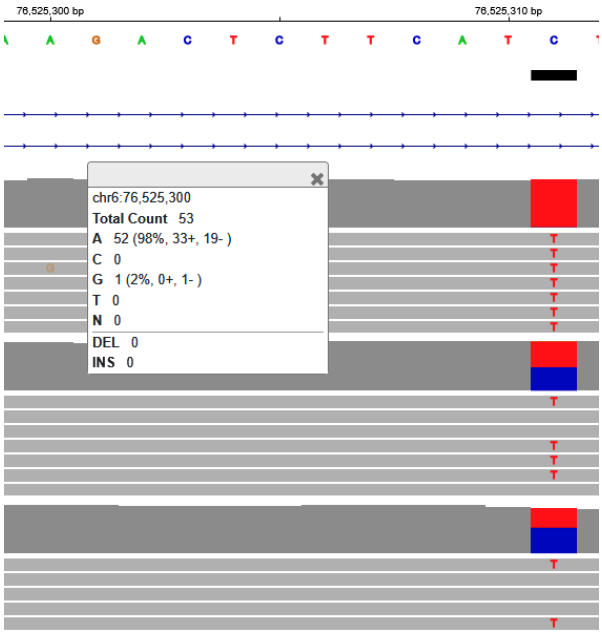
- Where a base in a read is the same as the reference, it is coloured grey
- Where a base in a read is different to the reference, it is coloured by base.



Zooming in even closer we can see the actual sequences and identify any differences between the reads and the reference.



To gauge whether any sequence differences might represent germline variants, we can examine the bases observed at each position by clicking on a base in the mountain plot at the top of each read track.



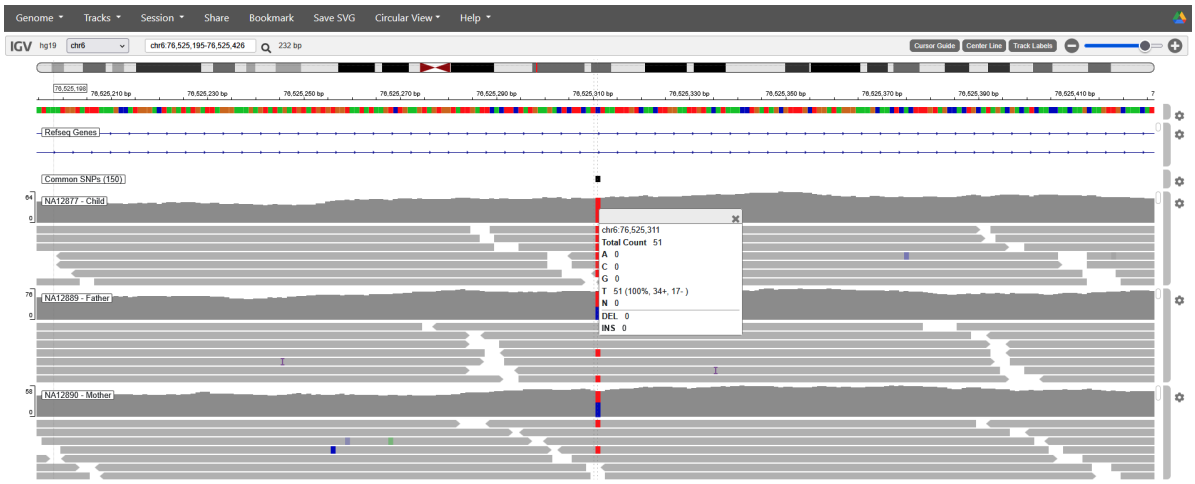
Test your understanding

- What do you think is the germline genotype for NA12877 (the son) at position ch6:76,525,300?
- What are the possible sources of the G read observed?

Answer

- This position appears to be homozygous A (A/A) because all the reads are the same
- Most likely a sequencing error, but could possibly be a rare somatic variant

Moving along slightly to position 76,525,311:



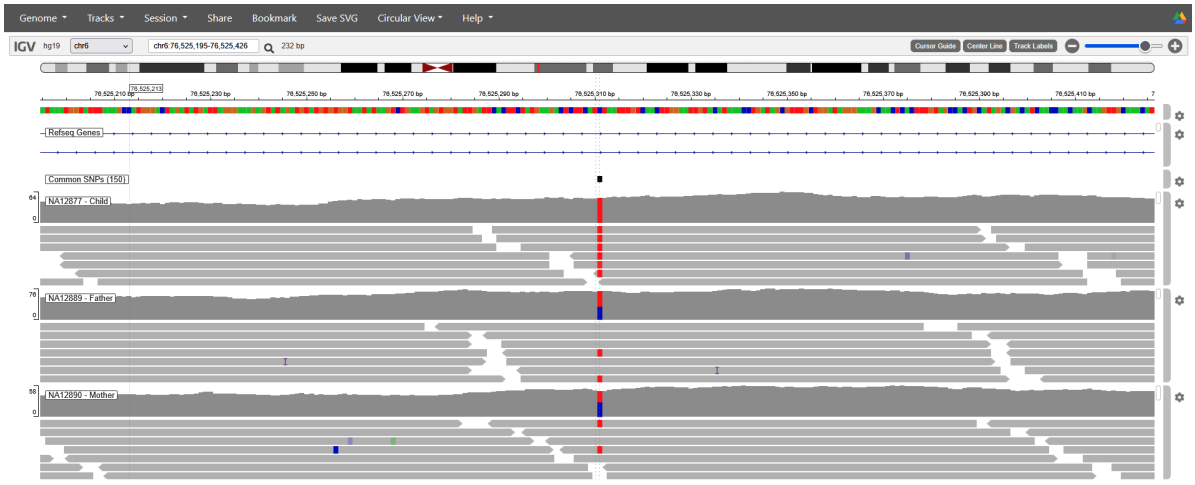
Test your understanding

- What is the reference sequence at position chr6:76,525,311?
- What is the germline genotype at position chr6:76,525,311 for NA12877 (the son)?
- How sure are we that this is not a sequencing error?

Answer

- The reference sequence at position chr6:76,525,311 is **C**
- The germline genotype at position chr6:76,525,311 for NA12877 is Homozygous T (T/T)
- All 51 reads contained T at this position, making a sequencing error highly unlikely. The germline genotype is therefore very likely to be T/T

Examine the reads for the other members of the trio.



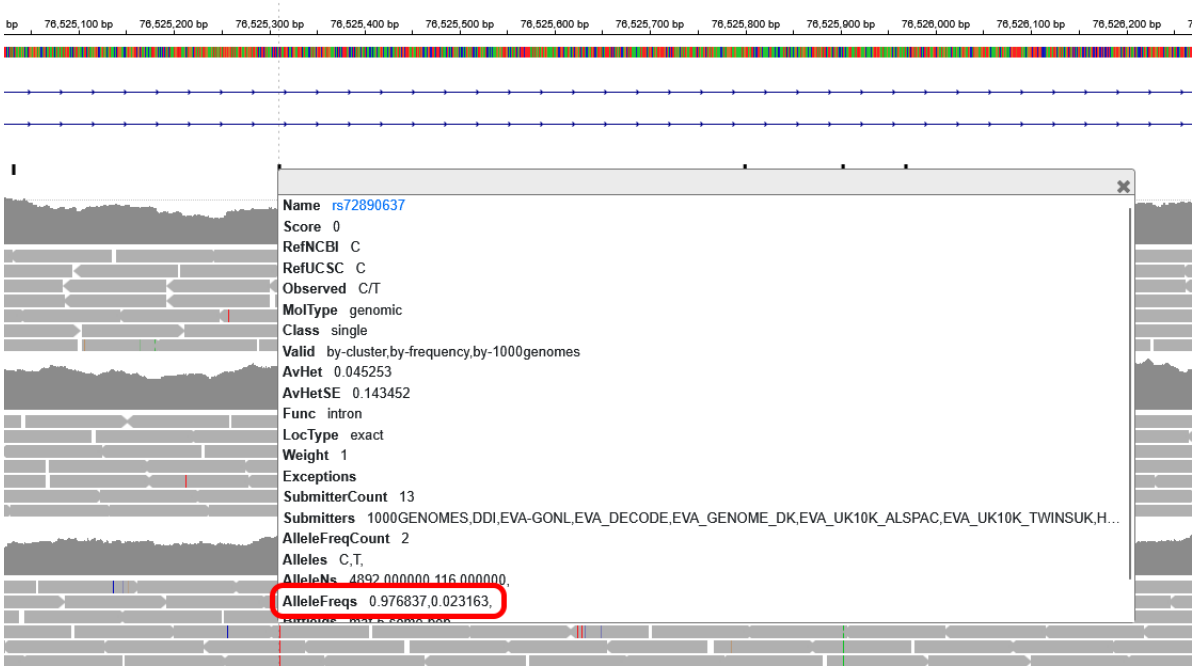
Test your understanding

- What are the genotypes at position chr6:76,525,311 for NA12889 and NA12890?
- Are the predicted genotypes consistent with Mendelian expectations?

Answer

- Both parents are heterozygous T/C
- The pattern does fit with Mendelian expectations: father T/C, mother T/C and son T/T suggest the son inherited a T from each parent

By clicking on the feature in the Common SNPs track we can see that this is a known SNP and gain additional information:



Test your understanding

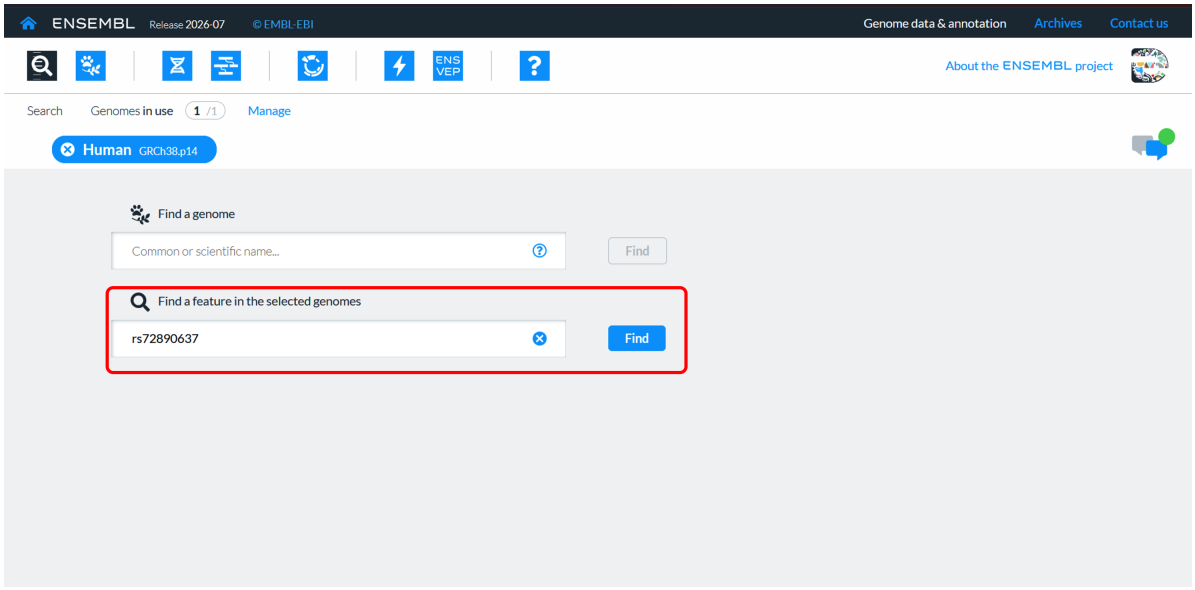
- What is the identifier (name) of the SNP at this position?
- What is the minor allele frequency (MAF)?

Answer

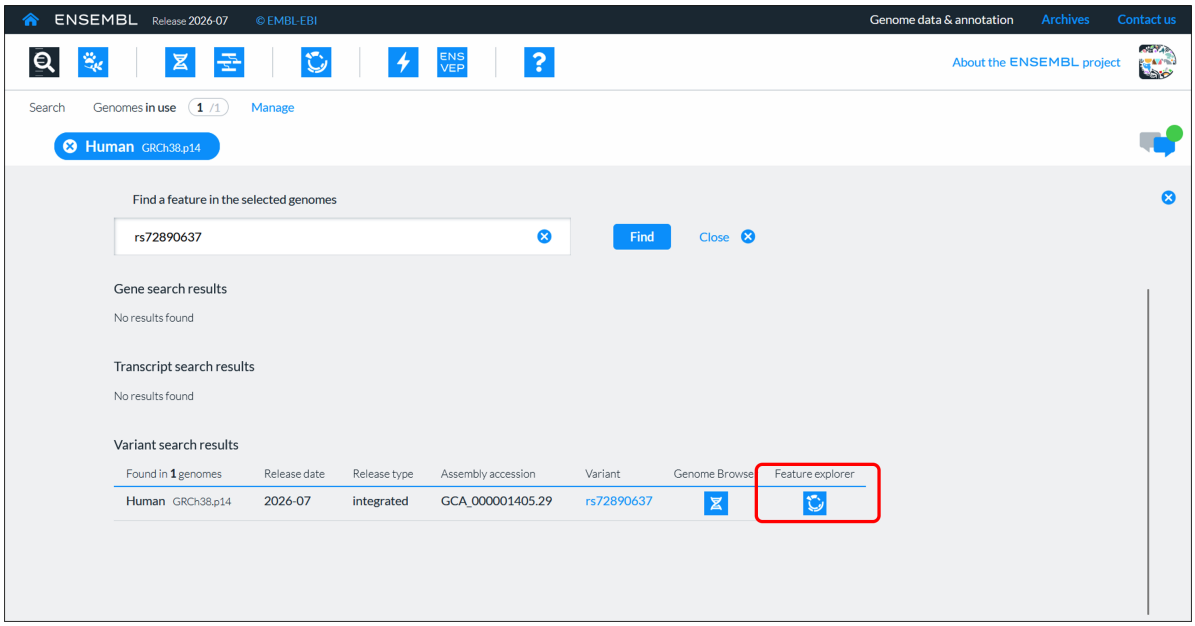
- The SNP identifier is rs72890637
- The MAF is 0.023

8 Investigate variants using Ensembl

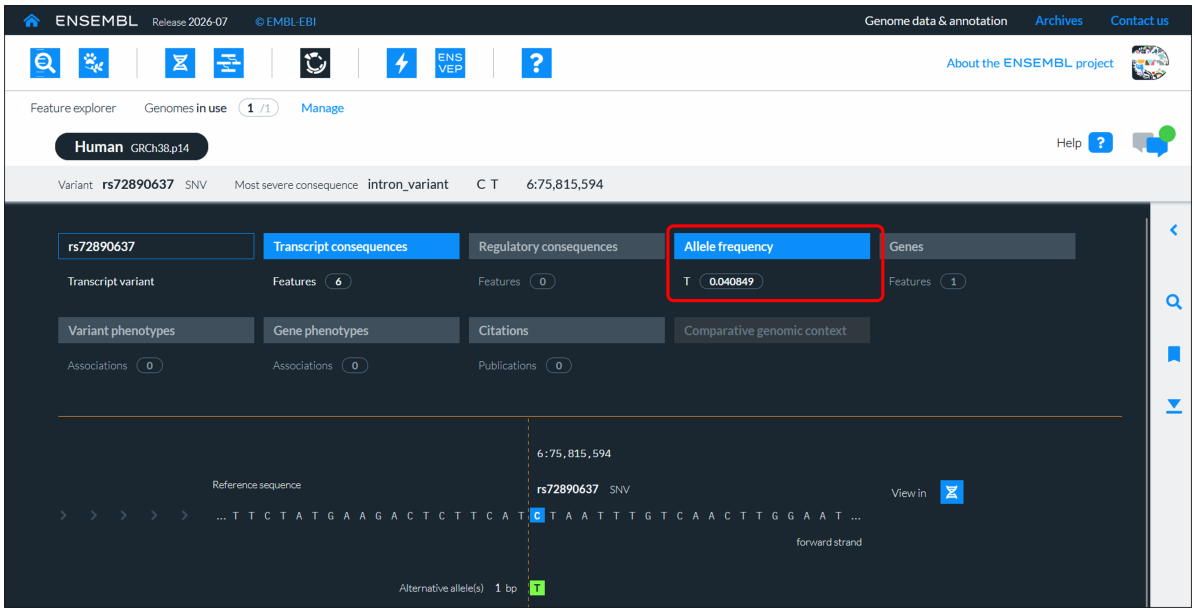
We can find out even more about this SNP using the Ensembl database. Enter the SNP identifier (rs72890637) into the Ensembl search box (If Ensembl search is down, use the direct link to the SNP)



Then click on the link in the Ensembl **Feature explorer** search results.



There are then links to various pieces of data about the SNP. Click on the **Allele Frequency** link.



Test your understanding

- What is the minor allele frequency in Finland? (you may have to expand the gnomAD genomes v4.1 dataset)

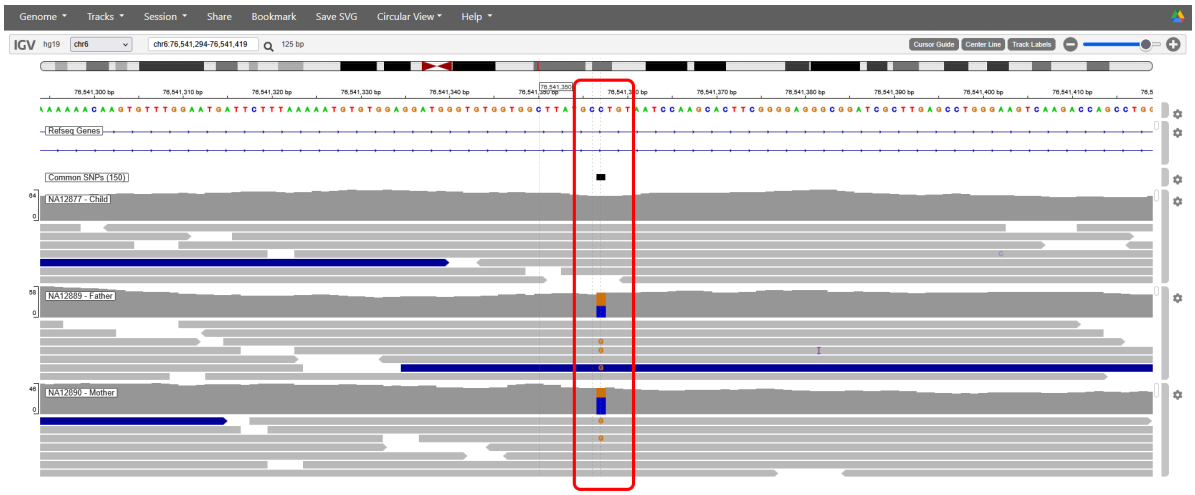
- How does that compare to the wider, non-Finnish European population?

Answer

- The MAF for Finnish Europeans is 9%
- The wider European population MAF is 6%. Finland often differs from other European populations because of a phenomenon called the Finnish founder effect that has emerged because of a small founder population, various bottlenecks, geographic isolation and low immigration.

9 Genotypes, inheritance and haplotypes

Let's look at a few more variants. To start with, we'll focus on rs1280055. We can jump straight to this SNP by typing its ID into the IGV search bar.



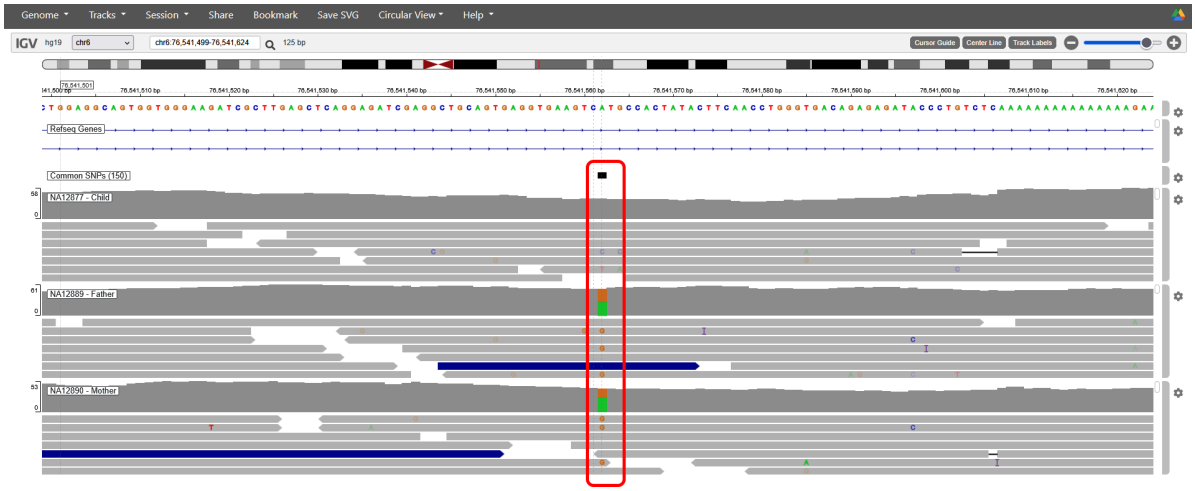
Test your understanding

- What are the genotypes at rs1280055 for NA12877, NA12889 and NA12890?
- Are they consistent with Mendelian inheritance patterns?

Answer

- The genotypes are C/C for NA12877 (child), G/C for NA12889 (father) and G/C for NA12890 (mother)
- These are consistent with Mendelian inheritance and the child has inherited a C allele from each heterozygous parent

Scroll along a few hundred bases to the right and you should find another SNP - rs1280056.



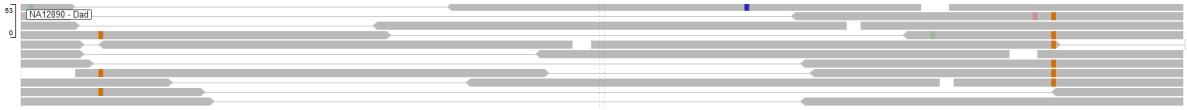
Test your understanding

- What are the genotypes at rs1280056 for NA12877, NA12889 and NA12890?

Answer

The genotypes are A/A for NA12877 (child), A/G for NA12889 (father) and A/G for NA12890 (mother)

To determine which variants have been inherited together on the same chromosome, and hence which parents they likely came from, we can 'phase' the sequence to determine haplotypes. In the example below, a G at rs1280055 is always found with a G in rs1280056 in mate pairs that cover both SNPs. Conversely, a C in rs1280055 is always found with an A in rs1280056. This allows us to infer the haplotype for the sample.

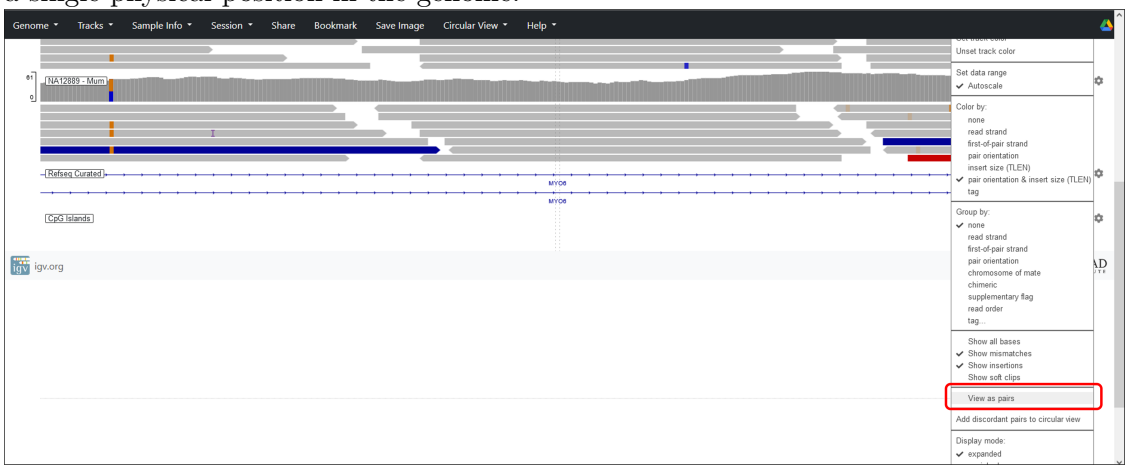


By using paired reads, we can look at much longer sequence distances than the 100 bp length of each individual read.



Tip

To help, we can use "view as pairs" in the parents to look for direct physical evidence of the parental haplotypes. This setting adds a line between read-pairs in the sequencing data, allowing us to see sequence that comes from the same fragment and originated from a single physical position in the genome.



Test your understanding

Use the family data and any read-pairs to phase the variants in the table and determine the haplotypes for rs1280056 and rs1280055 in NA12877, NA12889 and NA12890 (Don't forget to scroll up and down to view more reads). You may have to zoom to get both SNPs on screen at the same time.

Table 9.1: Genotypes and inferred haplotypes at rs1280056 and rs1280055.

Sample	Result	rs1280056	rs1280055
	Position	chr6:76,541,531–76,541,598	chr6:76,540,356–76,542,357
	Reference allele		
	Alternative allele		
NA12889 (father)	**Genotype**		
	Haplotype 1		
	Haplotype 2		
NA12890 (mother)	**Genotype**		
	Haplotype 1		

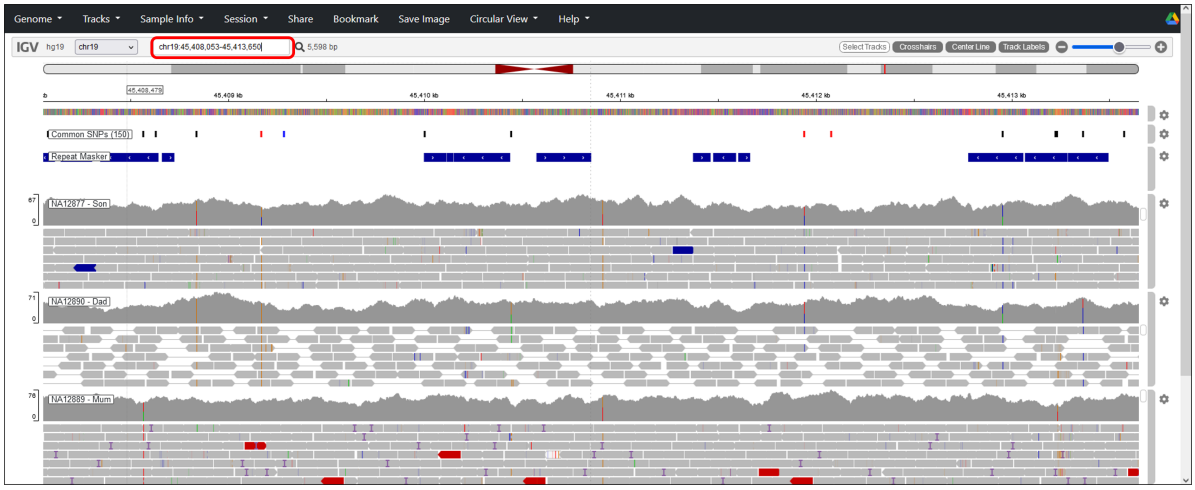
	Haplotype 2		
NA12877 (son)	**Genotype**		
	Haplotype 1		
	Haplotype 2		

Answer

Sample	Result	rs1280056	rs1280055
	Position	chr6:76,541,531–76,541,598	chr6:76,540,356–76,542,357
	Reference allele	A	C
	Alternative allele	G	G
NA12889 (father)	**Genotype**	A/G	C/G
	Haplotype 1	A	C
	Haplotype 2	G	G
NA12890 (mother)	**Genotype**	A/G	C/G
	Haplotype 1	A	C
	Haplotype 2	G	G
NA12877 (son)	**Genotype**	A/A	C/C
	Haplotype 1	A	C
	Haplotype 2	A	C

10 Variant effect predictor with *APOE*

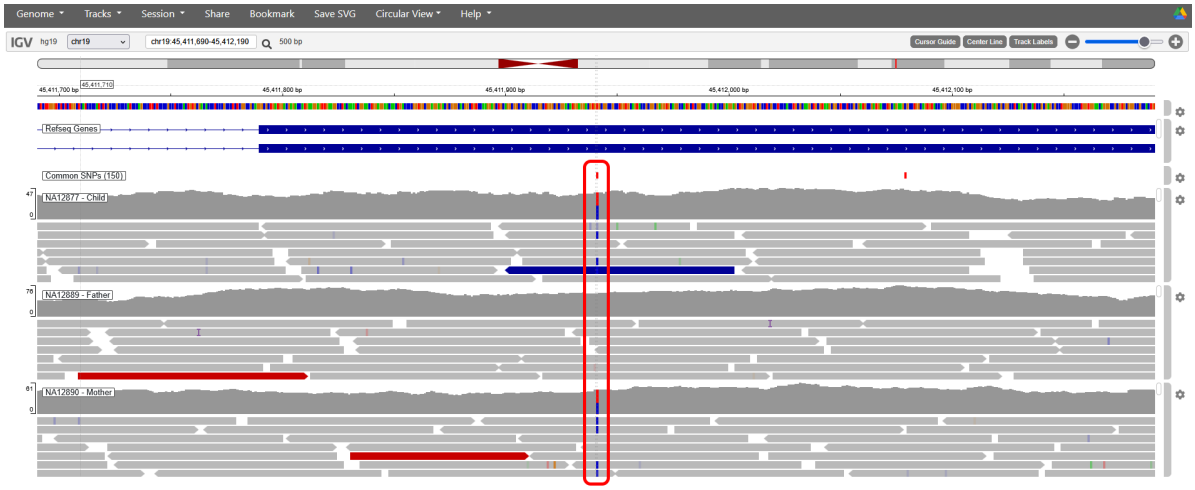
Next, we'll look at the *APOE* gene. You can quickly locate it using the search box.



Focus on the two common SNPs found in exon 4.



Start with the variant at position chr19:45,411,941 (rs429358).



Test your understanding

- What is the nature of the polymorphism at the DNA level?
- Does this variant change the amino acid sequence? (You will have to turn on translation and work out which is the reading frame by looking at exon 1)
- If so, from what to what?
- What are the genotypes at this SNP for NA12877, NA12889 and NA12890?
- What is the minor (alt) allele and the MAF?
- What type of variant is this – would it impact the protein?

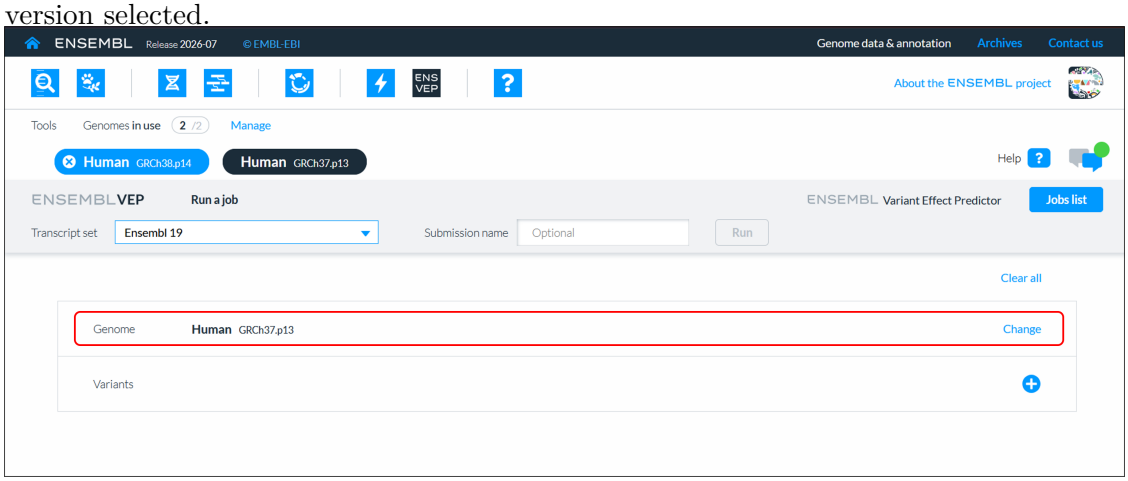
Answer

- The polymorphism at this site is C/T
- Yes, this is a nonsynonymous mutation
- The codon and amino acid changes from TGC (Cys) to CGC (Arg)
- The genotypes are C/T for NA12877 (child), T/T for NA12889 (father) and C/T for NA12890 (mother)
- The MAF for the minor (alt) C-allele is 18%
- This missense variant is protein altering. It is associated with predisposition to late-onset Alzheimer's disease (amongst other associations)

We can use the **variant effect predictor (VEP)** tool to gain some insight into potential functional effects of a given variant. Navigate to the VEP site: https://www.ensembl.org/tools/vep

Warning

We will be using hg19/GRCh37 coordinates, so make sure you have the right genome version selected.

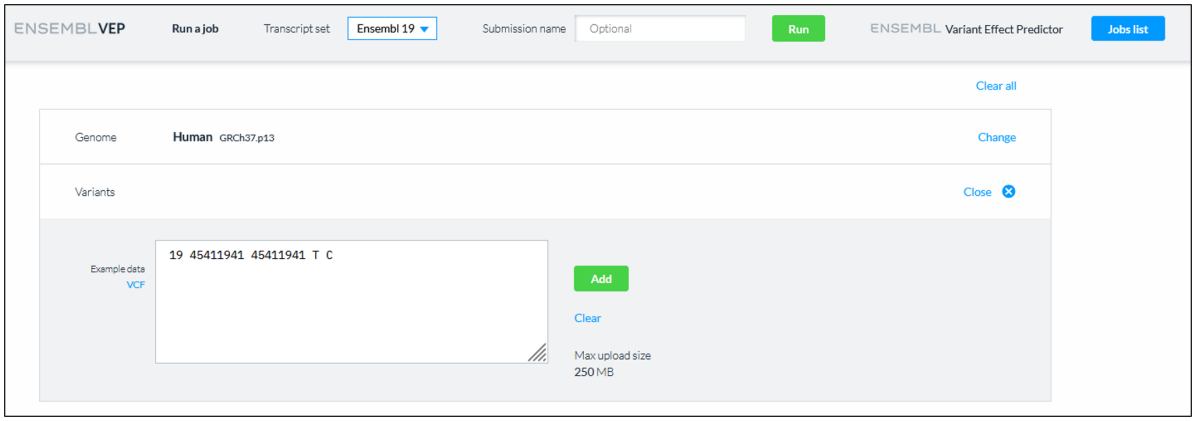


A VEP search string has to be in a set format:

```
19 45411941 45411941 T C
```

- The first number is the **Chromosome**
- The second number is the start position
- The third number is the end position (for a SNP, this is the same as the start position)
- The fourth character is the wild-type base
- The fifth character is the variant base(s) that the position will change to

Press "Add" when you've entered the details of the variant. Set the **submission name** to something you will remember.



We can also add some additional analyses to the job by scrolling down to **Job Options**. For now, enable **REVEL** and **CADD** under **Variant impact predictions**. These will help us predict any effect that the variants might have.

Tip

REVEL (Rare Exome Variant Ensemble Learner) is a computational tool that predicts whether a **missense** variant is likely to be disease-causing by combining the results of multiple variant effect prediction algorithms into a single score between 0 and 1. Higher REVEL scores indicate a greater likelihood that the variant is pathogenic, while lower scores suggest it is more likely to be benign. Although there are no set thresholds, a general rule of thumb is:

- 0.8+ : Pathogenic
- 0.5-0.8 : Likely Pathogenic
- 0.2-0.5 : Likely benign
- <0.2 : Benign

(Ioannidis NM, Rothstein JH, Pejaver V, et al. (2016) "REVEL: An Ensemble Method for Predicting the Pathogenicity of Rare Missense Variants." *American Journal of Human Genetics* 99(4):877-885. DOI:10.1016/j.ajhg.2016.08.016)

CADD (Combined Annotation Dependent Depletion) is a variant prioritisation score that integrates multiple genomic annotations to predict the deleteriousness of genetic variants. Higher CADD PHRED scores indicate a greater likelihood of functional impact, with scores above 20 typically considered potentially damaging.
CADD scores are ranked so that a score of 10 places a variant among the top 10% most deleterious variants in the genome, while a score of 20 places it among the top 1%.
Although there are no set thresholds for CADD, a general rule of thumb is:

- 30+ : Deleterious
- 20-30 : Potentially deleterious
- 10-20 : Unlikely to be deleterious
- <10 : Not deleterious

(Rentzsch P, Witten D, Cooper GM, Shendure J, Kircher M. (2019) "CADD: predicting the deleteriousness of variants throughout the human genome." *Nucleic Acids Research* 47(D1):D886-D894. DOI:10.1093/nar/gky1016)

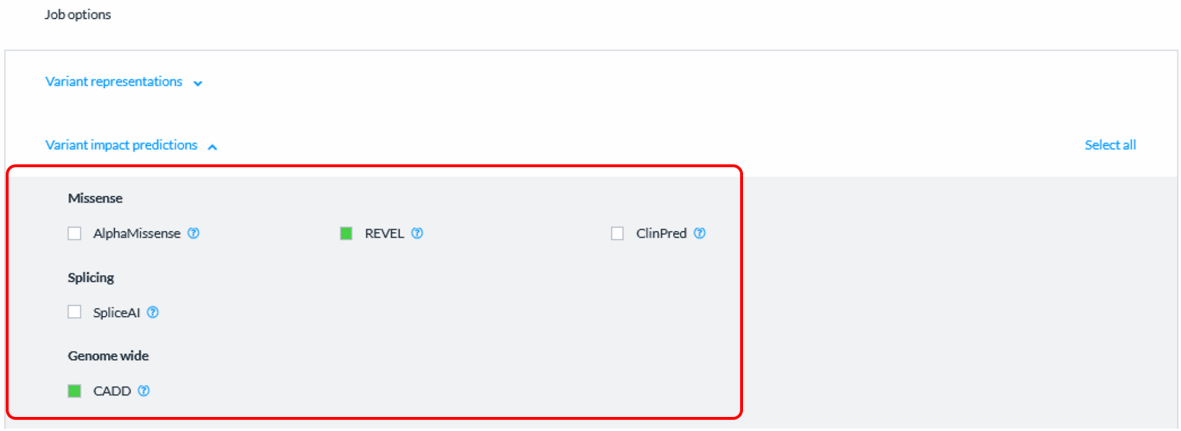
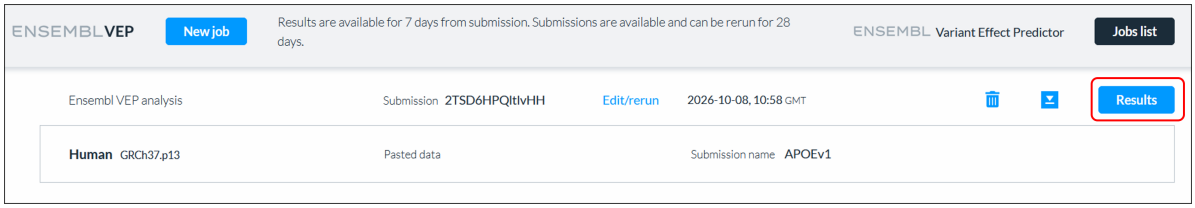


Figure 10.1: Screenshot from the original tutorial (slide 56).

Press "Run" when you're ready. Your job will take a few minutes. Press **Results** when it has finished.



Test your understanding

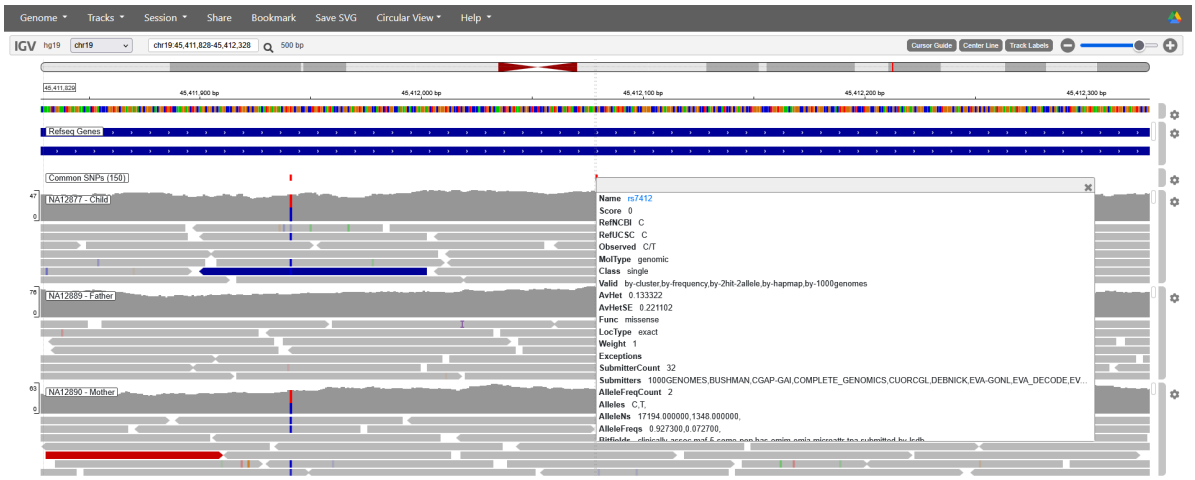
- What type of variant is this?
- What might the REVEL score indicate about this variant? (You might have to click show annotations to see these details)
- What might the CADD score indicate about this variant? (You might have to click show annotations to see these details)

Answer

- This is a missense variant
- A REVEL score of 0.229 is low and would generally be interpreted as evidence that the variant is likely benign (not disease-causing).
- A CADD PHRED score of 13.75 suggests the variant is unlikely to be deleterious.

These low scores may partly reflect the high allele frequency of approximately 15%. The variant is unlikely to be associated with a highly penetrant, early-onset condition. However, Alzheimer's disease is common and usually develops after reproductive age.

There is another SNP a bit further downstream (rs7412) we will examine next.



Test your understanding

- What is the nature of the polymorphism at the DNA level?
- Does this variant change the amino acid sequence?
- If so, from what to what?
- What are the genotypes at this SNP for NA12877, NA12889 and NA12890?
- Is this variant likely to be pathogenic? (check using VEP)

Answer

- The polymorphism is C/T
- Yes, this is a nonsynonymous mutation
- The codon and amino acid changes from CGC (Arg) to (TGC) Cys
- The variant has a higher REVEL score of 0.588 (use `19 45412079 45412079 C T` to search)

11 Case Study: Alzheimer's disease

Genetic, environmental and lifestyle factors contribute to late-onset Alzheimer's disease (late-onset AD, LOAD), with genetic factors contributing approximately 60%. Variation at the *APOE* gene locus plays a contributory genetic role. *APOE* encodes a very low-density lipoprotein that helps remove cholesterol from the bloodstream. The exact functional role of the protein encoded by *APOE* in LOAD remains unclear.

The major *APOE* risk for LOAD is generally assumed to derive from the 2/ 3/ 4 haplotype system, with the e4 allele increasing risk for the disorder and the e2 allele being protective. The 2/ 3/ 4 haplotype system is defined by the two non-synonymous SNPs in *APOE* exon 4 that we just examined. One is a C/T SNP (rs429358) that encodes either cysteine (C) or arginine (R) at amino acid 130. The second site defining this haplotype system is a C/T SNP (rs7412), which again encodes arginine (R) or cysteine (C) at amino acid 176.

The allelic compositions of the commonly investigated rs429358/rs7412 haplotypes are:

- T–T for 2
- T–C for 3
- C–C for 4

	rs429358	rs7412	Frequency and Prognosis
APOE 2	T	T	Rare, possibly protective
APOE 3	T	C	Common, neutral
APOE 4	C	C	20% frequency, risk variant
Not observed	C	T	

Test your understanding

- What are the *APOE* exon 4 haplotypes for NA12877, NA12889 and NA12890 and their relative LOAD risk status?
- Why might the C–T haplotype not be observed?

Answer

The haplotypes and relative LOAD risk are:

- NA12877: T–T *APOE* 2 possibly protective, T–C *APOE* 3 neutral
- NA12889: T–T *APOE* 2 possibly protective, T–T *APOE* 2 possibly protective
- NA12890: T–T *APOE* 2 possibly protective, T–C *APOE* 3 neutral

The C-T haplotype may not be observed, as the two variants are very close and the probability of a recombination event generating the fourth recombinant haplotype is very low

12 Case Study: Zellweger syndrome

A family is referred to clinical genetics with a severely ill newborn child who has received a clinical diagnosis of probable Zellweger syndrome. This is an autosomal recessive systemic disorder characterised clinically by severe neurologic dysfunction, craniofacial abnormalities, and liver dysfunction, and biochemically by the absence of peroxisomes. Zellweger syndrome is characterised by locus heterogeneity and can be caused by mutations in any one of several genes, encoding proteins known as pexins, which are involved in peroxisome biogenesis. In order to try and provide a genetic diagnosis, whole exome sequencing was performed in the family and variants in a known Zellweger gene, *PEX3*, detected:

Gene	Position (GRCh37/hg19)	Mother	Father	Affected child
PEX3	6:143784092	A/A	A/G	A/A
PEX3	6:143792714	G/TG	G/G	G/TG
PEX3	6:143795928	T/T	T/A	T/A
PEX3	6:143800237	T/C	T/T	T/T

Test your understanding

- How would you format a VEP query for each variant?
- What are the predicted effects of these variants?

Answer

The VEP search strings and likely effects would be:

- `6 143784092 143784092 A G` - missense (Q/R), likely benign
- `6 143792714 143792714 G TG` - frameshift, likely deleterious
- `6 143795928 143795928 T A` - stop-gained variant, predicted to be deleterious
- `6 143800237 143800237 T C` - benign

You can investigate the variants further in gnomAD. Make sure to use the gnomAD version 2.1.1 so that GRCh37/hg19 is used and matches the genome coordinates you have been using in IGV. Complete the following table:

Test your understanding

Complete the table:

Gene	Position	Genotype	Observed	Global allele frequency	Number of homozygotes
PEX3	6:143784092				
PEX3	6:143792714				
PEX3	6:143795928				
PEX3	6:143800237				

How would you interpret these data in terms of providing a diagnosis and future risk assessment to the family? (In answering this question, remember that the impact of a variant on the gene product may or may not correlate with the effect size in terms of downstream phenotype. You'll therefore need to use all the evidence available to you to predict whether variants you are seeing in the data might be causative of Zellweger syndrome or not)

Answer

Gene	Position	Genotype	Observed	Global allele frequency	Number of homozygotes
PEX3	6:143784092	A/G	yes (rs35220041)	0.02818	615
PEX3	6:143792714	G/TG	no	N/A (<1/~282,000)	0
PEX3	6:143795928	T/A	no	N/A (<1/~251,000)	0
PEX3	6:143800237	T/C	no	N/A (<1/~251,000)	0

The data strongly support a diagnosis of Zellweger syndrome in the affected child, who is compound heterozygous for two predicted high-impact PEX3 variants, one frameshift and one premature stop variant, neither of which has been observed in gnomAD. Although the impact of the 143784092 variant is predicted as moderate, this allele is relatively frequent (~2% globally and 17% in the Latino/admixed American population), and at least 600 homozygotes have been observed, so it is unlikely to be a disease allele. The 143800237 variant is novel, but synonymous, so it is again unlikely to be pathogenic. Thus, each parent carries a different candidate pathogenic allele. The recurrence risk is 25% for each pregnancy, assuming the variants are confirmed to be in trans and causative.

13 References and useful links

- Eberle MA et al. (2017). A reference data set of 5.4 million phased human variants validated by genetic inheritance from sequencing a three-generation 17-member pedigree. *Genome Research* 27:157–164. https://doi.org/10.1101/gr.210500.116
- IGV web application: https://smb-informatics.mvls.gla.ac.uk/igv
- University VPN: https://www.gla.ac.uk/myglasgow/it/vpn/
- Ensembl: https://www.ensembl.org/
- Ensembl VEP: https://www.ensembl.org/tools/vep
- gnomAD: https://gnomad.broadinstitute.org/